

SANJANA NAMBIAR

☎ +971 56 945 4313 ✉ sanjana.nambiar04@gmail.com 🌐 [sanjanavnambiar](#) 🌐 [Sanjana-Nambiar](#) 🌐 [Portfolio](#)

EDUCATION

Bachelor of Science in Computer Science, New York University Abu Dhabi **Aug 2021 – May 2025**
Minors: Applied Mathematics and Engineering GPA: 3.7
Relevant Coursework: Deep Learning and LLM-based Generative AI Systems(Graduate level), Processing Big Data for Analytics Applications, Software Engineering, Applied Machine Learning
CBSE Higher Secondary (AISSCE), Sharjah Indian School **Apr 2019 - Mar 2021**
Ranked 3rd school-wide, topped Engineering Graphics and Chemistry. GPA: 3.9

SKILLS

Programming Languages	C++ (4 yrs), Python (4 yrs), C (2 yrs), JavaScript (2 yrs), MATLAB (1 yr)
Machine Learning	PyTorch, TensorFlow, Keras, Hugging Face, Transfer Learning, Fine-tuning (LoRA), Synthetic Dataset Creation, Parameter Optimization (WandB)
Web Development	Node.js, Express.js, HTML5, CSS3, Flask, PostgreSQL, MySQL, MongoDB, Gunicorn, Nginx, Firebase, Flutter, Dart
Advanced Computing	Expertise in High-Performance Computing (HPC), Cluster Computing, Slurm Scheduler, Parallel Programming, Efficient Memory Management, Large Dataset Handling
Hardware Skills	Raspberry Pi, Arduino, Soldering, 3D Printing, ROS Noetic
Languages	English (Proficient), Hindi (Proficient), Malayalam (Native), Arabic (Beginner)

WORK EXPERIENCES

Junior Research Scientist **Feb 2026 – Present**
Center for Interacting Urban Networks (CITIES) Research Institute Abu Dhabi, UAE
Advisors: Prof. Azza Abouzied & Prof. Christina Pöpper

- Red-teaming frontier LLMs to identify vulnerabilities in handling data-void topics and misinformation.
- Designing adversarial prompts and developing data-void evaluation pipelines to probe model behavior in low-information domains and analyze failure modes such as hallucinations and misleading claims.

AI Product Researcher **Aug 2025 – Sep 2025**
Trevex Dubai, UAE

- Engineered GenAI-driven agentic workflows in credit/risk management using n8n, LangChain, ElevenLabs API, and Google Cloud Auth; automated case classification, NDA processing, and client outreach, cutting manual effort by 75%.
- Built and deployed a Conversational AI agent integrating text-to-speech, calendar APIs, and automated reminders, reducing average client response latency by 60% and autonomously scheduling follow-ups weekly.

Software Engineer (Analytics Dashboard & DevOps) **May 2024 - Aug 2024**
Letsrise Abu Dhabi, UAE

- Designed and implemented a scalable user analytics and admin dashboard using Figma, Flask, and PostgreSQL, handling data pipelines for 250+ user datasets to ensure robust insights on entrepreneurial traits.
- Automated data pipelines using PostgreSQL for real-time user data processing, ensuring system performance and scalability.
- Developed and deployed the dashboard using Flask, Nginx, Gunicorn, and Cloudflare, optimizing system performance by 25%, resulting in successful investor pitches and onboarding four new users.

RESEARCH EXPERIENCES

Research Assistant (AI/LLM Security) **Feb 2024 - Jul 2025**
Cyber Security and Privacy (CSP) Lab, Center for Cyber Security Abu Dhabi (CCSAD) Abu Dhabi, UAE
Advisor: Prof. Christina Pöpper

- **PPTP:** Explored transformer hidden states as early indicators of malicious prompt intent. Developed a classifier using LLaMA-2-7B activations achieving 94.5% accuracy and deployed it as a runtime guardrail pipeline.
- **Capstone:** Created JailFact-Bench, a 99-prompt dataset comparing jailbreak and factual prompts by curating, refining, and validating paired inputs, enabling comprehensive analysis of LLM vulnerabilities.
- Uncovered a 25–40% factual drift in jailbreak outputs versus factual prompts by conducting statistical analysis on semantic similarity and human-annotated accuracy, redefining risk metrics for adversarial inputs.
- Automated data pipelines and toxicity analysis using Python and APIs (OpenAI Moderation, Perspective), delivering actionable insights to improve LLM alignment and mitigate hallucinations.

Research Assistant (LLM Alignment & Robustness)

Data Intelligence and Computation in Engineering (DICE) Lab, Tandon School of Engineering, NYU

Jul 2024 - Jan 2025

New York, US

Advisor: Prof. Chinmay Hegde

- Researched on defense strategies for AI models against Jailbreak attacks, achieving a significant reduction in attack success rates from 71.42% to 12.24% through synthetic fine-tuning with counter-fake PII datasets.
- Co-authored a paper presented at ICLR 2025, introducing a benchmark to evaluate biases in LLM judges across alignment metrics like helpfulness, honesty, and harmlessness.
- Discovered and analyzed reference stuffing as an injection attack vector, reducing LLM performance in preference ranking systems by 49%.

Research Assistant (IoT Environmental Station)

Mubadala Arabian Center for Climate and Environmental Sciences (ACCESS) Lab

Feb 2023 - May 2024

Abu Dhabi, UAE

Advisor: Prof. Francesco Paparella

- Designed and deployed 3 IoT-based Environmental Monitoring Stations with Raspberry Pi 4 and advanced sensors (BME280, SCD30, NextPM), achieving 95% data accuracy and reducing assembly time from one month to one day through workflow optimization and 3D-printed sensor bases.
- Implemented real-time monitoring in remote areas by integrating cellular connectivity, reverse tunneling (ngrok), and automated data collection processes using shell scripts.
- Conducted rigorous Python-based testing and debugging to ensure precise, reliable data acquisition, enhancing system scalability and operational efficiency for environmental research.

RESEARCH & PUBLICATIONS

JailFact-Bench: A Comprehensive Analysis of Jailbreak Attacks vs. Hallucinations in LLMs

Jun 2025

Sanjana Nambiar, & Christina Pöpper

Published in *SiMLA 2025 (7th ACNS Workshops)*, Springer LNCS, 2025. [\[Springer\]](#) [\[PDF\]](#) [\[Dataset\]](#) [\[Slides\]](#)

Style Over Substance: Failure Modes of LLM Judges in Alignment Benchmarking

Apr 2025

Benjamin Feuer, Micah Goldblum, Teresa Datta, **Sanjana Nambiar**, Raz Besaleli, Samuel Dooley, Max Cembalest, John P.

Published at *ICLR 2025*. Poster presentation. [\[PDF\]](#)

AWARDS & CERTIFICATIONS

Technical AI safety Course – Bluedot Impact [\[Link\]](#)

Jan 2026

- Completed a 30-hour program on AI safety, threat modeling, and kill chain analysis; developed an actionable plan to contribute to safer AI systems.

Highly Commended – Center for Urban Science and Progress Data Dive [\[Link\]](#)

Feb 2024

- Analyzed cycling & walking's impact on air quality; proposed mobility policies based on pollutant exposure in dense areas.

Second Place – NYUAD Hackathon for Social Good (Team Qatrah) [\[Code\]](#)

Apr 2023

- Built a quantum-enhanced water distribution system using NetworkX and QUBO for optimized sensor placement.

Finalist – CSAW'22 Cybersecurity Games and Conference (Hack My Robot) [\[Link\]](#)

Nov 2022

- Built & tested a ROS-Noetic TurtleBot3. Conducted DoS attacks with Kali Linux; visualized motion tracking using RViz.

Super Achiever – Middle East Education Award [\[Link\]](#)

Aug 2021

- Recognized among top performers at the 7th India Middle East Education Awards with a 96.2% Grade 12 score.

LEADERSHIP & COMMUNITY ENGAGEMENT

Volunteer – 12th Annual International Hackathon for Social Good

May 2024 – Jun 2024

- Supported logistics and participant coordination for 180 students from 50+ countries, ensuring a seamless event experience.

Events Board Member – Undergraduate Student Government

Feb 2023 – May 2024

- Led campus-wide events (e.g. 2024 Gala, Valentine's Day), engaging 1,000+ students and enhancing community spirit.

Communications Officer – Melting Pot

Sep 2022 – Jun 2023

- Streamlined event logistics and communications, boosting outreach and cross-department coordination.

Resources Core Team Member – weSTEM (Women Empowered in STEM)

Mar 2022 – Dec 2022

- Co-developed a resource guide for 500+ CS and Math undergrads, integrating alumni insights on career planning.

Committee Member – Sustainability Committee, Student Government

Feb 2022 – Jun 2022

- Drove sustainability initiatives through campus-wide surveys, presented green program proposals to university leadership.